The reduced stated skein algebra of the once-punctured bigon and the Drinfeld–Tanisaki double

Conventions

Let q be an indeterminate and let

$$\Bbbk = \mathbb{Q}(q^{1/n}).$$

All algebras are over $\Bbbk$. We use the standard simply-connected form of $\mathcal{U}_q(\mathfrak{sl}_n)$, with toral generators $\widehat{K}_1, \dots, \widehat{K}_n$ satisfying $\widehat{K}_1 \cdots \widehat{K}_n = 1$. Put

$$K_i = \widehat{K}_i \widehat{K}_{i+1}^{-1}, \qquad 1 \le i \le n-1.$$

Let $\mathcal{U}_q(\mathfrak{b}_+)$ be generated by E_i and the $\widehat{K}_j^{\pm 1}$, and $\mathcal{U}_q(\mathfrak{b}_-)$ by F_i and the $\widehat{K}_j^{\pm 1}$. The skew pairing is the Drinfeld–Tanisaki pairing

$$\langle \widehat{K}_i, \widehat{K}_j \rangle = q^{\frac{1}{n} - \delta_{ij}}, \qquad \langle E_i, F_j \rangle = \frac{\delta_{ij}}{q^{-1} - q}, \qquad \langle \widehat{K}_i, F_j \rangle = \langle E_j, \widehat{K}_i \rangle = 0.$$

We denote by

$$D := \mathcal{U}_q(\mathfrak{b}_-) \bowtie \mathcal{U}_q(\mathfrak{b}_+)$$

the corresponding generalized quantum double.

Let $\mathfrak{B}^*$ be the once-punctured bigon. The symbol $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$ denotes the *reduced* stated SL_n-skein algebra. Let b_{ij} and c_{ij} denote the two standard horizontal arcs, both oriented from left to right, with states i at the left endpoint and j at the right endpoint. With the boundary order convention used below, the reduced relations give

$$b_{ij} = 0 \quad (i < j), \qquad c_{ij} = 0 \quad (i > j).$$

1 The FRT presentation

Let e_{ij} be the matrix units in $\mathrm{End}(\Bbbk^n)$, and let

$$R = q^{-1/n} \left(q \sum_i e_{ii} \otimes e_{ii} + \sum_{i \neq j} e_{ii} \otimes e_{jj} + (q - q^{-1}) \sum_{i<j} e_{ij} \otimes e_{ji} \right)$$

be the standard type A_{n-1} R-matrix. For a matrix $X = (x_{ij})$ with entries in an algebra, write $X_1 = X \otimes I$ and $X_2 = I \otimes X$.

Lemma 1.1 (Skein presentation of $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$)**.** *The algebra $\overline{\mathcal{S}}_n(\mathfrak{B}^*)$ is generated by the entries of the lower triangular matrix $B = (b_{ij})$ and the upper triangular matrix $C = (c_{ij})$, subject to*

$$RB_1B_2 = B_2B_1R, \qquad RC_1C_2 = C_2C_1R, \qquad RC_1B_2 = B_2C_1R,$$

together with

$$b_{ij} = 0 \ (i < j), \qquad c_{ij} = 0 \ (i > j), \qquad \prod_{i=1}^{n} b_{ii} = 1, \qquad \prod_{i=1}^{n} c_{ii} = 1.$$

In particular all b_{ii} and c_{ii} are invertible.

Proof. This is the standard reduced stated-skein presentation. The triangularity comes from the bad-arc reduction at the two boundary vertices. Each of the two subalgebras generated by one family of horizontal arcs is the reduced skein algebra of a once-punctured monogon, hence is the quantized coordinate algebra of the corresponding Borel subgroup; this gives the two RTT relations and the determinant-one relations. The mixed relation is the braided tensor-product relation obtained by gluing over the ideal triangle around the puncture; the coquasitriangular form on the bigon is precisely the FRT R-matrix form. See Lê–Sikora [3, Theorems 6.3, 6.4, 8.2] for the bigon Hopf algebra, its co-R-matrix and the gluing theorem, and Lê–Yu [4, Sections 3.6 and 5] for the reduced punctured monogon and the Borel quotient. □

Let $L^- = (\ell^-_{ij})$ and $L^+ = (\ell^+_{ij})$ be the FRT L-operators in the two Borel halves of the double D. Thus

$$\ell^-_{ij} = 0 \ (i < j), \qquad \ell^+_{ij} = 0 \ (i > j),$$

and, writing H_i^- and H_i^+ for the toral generators in the lower and upper Borel copies,

$$\ell^-_{ii} = (H_i^-)^{-1}, \qquad \ell^+_{ii} = H_i^+,$$
$$\ell^-_{i+1,i} = (q^{-1} - q)F_i(H_i^-)^{-1}, \qquad \ell^+_{i,i+1} = (q - q^{-1})H_i^+E_i.$$

The remaining entries are the standard higher root-vector entries of the FRT matrices.

Lemma 1.2 (FRT presentation of the double)**.** *The assignment*

$$b_{ij} \longmapsto \ell^-_{ij} \bowtie 1, \qquad c_{ij} \longmapsto 1 \bowtie \ell^+_{ij}$$

extends to an algebra isomorphism from the algebra in Lemma 1.1 onto $D = \mathcal{U}_q(\mathfrak{b}_-) \bowtie \mathcal{U}_q(\mathfrak{b}_+)$.

Proof. The L-operators satisfy the RLL relations

$$RL_1^-L_2^- = L_2^-L_1^-R, \qquad RL_1^+L_2^+ = L_2^+L_1^+R, \qquad RL_1^+L_2^- = L_2^-L_1^+R,$$

which are precisely the RTT and mixed RTT relations above. Moreover the entries generate the two Borel halves: indeed

$$H_i^+ = (\ell^+_{ii}), \qquad E_i = (q - q^{-1})^{-1}(\ell^+_{ii})^{-1}\ell^+_{i,i+1},$$

and

$$(H_i^-)^{-1} = \ell^-_{ii}, \qquad F_i = (q^{-1} - q)^{-1}\ell^-_{i+1,i}(\ell^-_{ii})^{-1}.$$

Thus the homomorphism is surjective.

Injectivity follows from the PBW theorem for the RLL presentation: ordered monomials in the lower root entries, the two torus copies, and the upper root entries form a basis; this basis is identified with the usual PBW basis of the Drinfeld–Tanisaki double. This is the standard FRT presentation of the quantum double of the two Borel subalgebras; see Klimyk–Schmüdgen [2, Chapters 8–10] or Chari–Pressley [1, Chapter 9]. □

2 The skein algebra as a quantum double

Theorem 2.1. *There is an explicit algebra isomorphism*

$$\Phi : \overline{\mathcal{S}}_n(\mathfrak{B}^*) \xrightarrow{\cong} \mathcal{U}_q(\mathfrak{b}_-) \bowtie \mathcal{U}_q(\mathfrak{b}_+)$$

given on the horizontal arcs by

$$\Phi(b_{ij}) = \ell^-_{ij} \bowtie 1, \qquad \Phi(c_{ij}) = 1 \bowtie \ell^+_{ij}.$$

Equivalently, on the simple generators,

$$\Phi(b_{ii}) = (H_i^-)^{-1} \bowtie 1, \qquad \Phi(c_{ii}) = 1 \bowtie H_i^+,$$
$$\Phi(b_{i+1,i}) = (q^{-1} - q)F_i(H_i^-)^{-1} \bowtie 1, \qquad \Phi(c_{i,i+1}) = 1 \bowtie (q - q^{-1})H_i^+ E_i.$$

The inverse sends

$$H_i^+ \longmapsto c_{ii}, \qquad E_i \longmapsto (q - q^{-1})^{-1} c_{ii}^{-1} c_{i,i+1},$$

and

$$(H_i^-)^{-1} \longmapsto b_{ii}, \qquad F_i \longmapsto (q^{-1} - q)^{-1} b_{i+1,i} b_{ii}^{-1}.$$

Proof. Combine Lemma 1.1 with Lemma 1.2. □

3 The quotient by $b_{ii}c_{ii} - 1$

Let

$$I = \langle b_{ii}c_{ii} - 1 \mid 1 \le i \le n \rangle \subset \overline{\mathcal{S}}_n(\mathfrak{B}^*)$$

be the two-sided ideal generated by the diagonal relations.

Lemma 3.1. *The quotient of $D = \mathcal{U}_q(\mathfrak{b}_-) \bowtie \mathcal{U}_q(\mathfrak{b}_+)$ by the ideal*

$$J = \left\langle (H_i^-)^{-1} H_i^+ - 1 \mid 1 \le i \le n \right\rangle$$

is canonically isomorphic to $\mathcal{U}_q(\mathfrak{sl}_n)$.

Proof. There is a natural surjective homomorphism

$$D \longrightarrow \mathcal{U}_q(\mathfrak{sl}_n)$$

sending the lower Borel copy to the usual lower Borel, the upper Borel copy to the usual upper Borel, and identifying the two torus copies:

$$H_i^- \longmapsto \widehat{K}_i, \qquad H_i^+ \longmapsto \widehat{K}_i, \qquad F_i \longmapsto F_i, \qquad E_i \longmapsto E_i.$$

The generators $(H_i^-)^{-1}H_i^+ - 1$ lie in its kernel, so the map factors through D/J.

The PBW decomposition of the double is

$$D \cong \mathcal{U}_q(\mathfrak{n}_-) \otimes \Bbbk[P] \otimes \Bbbk[P] \otimes \mathcal{U}_q(\mathfrak{n}_+),$$

where P is the weight lattice of $\mathfrak{sl}_n$. Modding out by J identifies the two copies of $\Bbbk[P]$, giving

$$D/J \cong \mathcal{U}_q(\mathfrak{n}_-) \otimes \Bbbk[P] \otimes \mathcal{U}_q(\mathfrak{n}_+)$$

as a vector space. This is exactly the PBW vector space of the simply-connected $\mathcal{U}_q(\mathfrak{sl}_n)$. Hence the induced surjection $D/J \to \mathcal{U}_q(\mathfrak{sl}_n)$ is an isomorphism. □

Theorem 3.2. *There is a canonical algebra isomorphism*

$$\overline{\mathcal{S}}_n(\mathfrak{B}^*)/I \;\cong\; \mathcal{U}_q(\mathfrak{sl}_n).$$

Under this isomorphism,

$$b_{ii} \longmapsto \widehat{K}_i^{-1}, \qquad c_{ii} \longmapsto \widehat{K}_i,$$
$$b_{i+1,i} \longmapsto (q^{-1}-q)F_i\widehat{K}_i^{-1}, \qquad c_{i,i+1} \longmapsto (q-q^{-1})\widehat{K}_iE_i.$$

Equivalently, in the quotient,

$$\widehat{K}_i \longmapsto c_{ii} = b_{ii}^{-1}, \qquad E_i \longmapsto (q-q^{-1})^{-1}c_{ii}^{-1}c_{i,i+1},$$

and

$$F_i \longmapsto (q^{-1}-q)^{-1}b_{i+1,i}b_{ii}^{-1}.$$

Proof. By Theorem 2.1, the ideal I corresponds exactly to the ideal J of Lemma 3.1, because

$$\Phi(b_{ii}c_{ii}) = \big((H_i^-)^{-1}\bowtie 1\big)\big(1\bowtie H_i^+\big) = (H_i^-)^{-1}\bowtie H_i^+.$$

Therefore

$$\overline{\mathcal{S}}_n(\mathfrak{B}^*)/I \cong D/J \cong \mathcal{U}_q(\mathfrak{sl}_n).$$

The displayed formulas are the images of the FRT $L^{\pm}$-entries after the torus identification $H_i^- = H_i^+ = \widehat{K}_i$. □

Remark 3.3. *Because $\prod_i b_{ii} = 1 = \prod_i c_{ii}$, the n relations $b_{ii}c_{ii} = 1$ contain one redundancy. Keeping all n generators of the ideal is nevertheless natural and symmetric.*

References

[1] V. Chari and A. Pressley, *A Guide to Quantum Groups*, Cambridge University Press, 1994.

[2] A. Klimyk and K. Schmüdgen, *Quantum Groups and Their Representations*, Springer, 1997.

[3] T. T. Q. Lê and A. S. Sikora, *Stated $SL(n)$-skein modules and algebras*, J. Topology 17 (2024), no. 3, e12350; arXiv:2201.00045.

[4] T. T. Q. Lê and T. Yu, *Quantum traces for SL_n-skein algebras*, arXiv:2303.08082.

[5] S. Majid, *Foundations of Quantum Group Theory*, Cambridge University Press, 1995.